

Emre Bulut

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PROFESSIONAL SUMMARY

Electrical and Electronics Engineering student working at the intersection of RF/microwave engineering and machine learning. I build signal-processing pipelines that turn raw software-defined radio captures into trainable datasets, and I train and deploy computer vision models on edge hardware — grounded in hands-on enterprise virtualization, Linux automation, embedded telemetry, and field electrical work. I am looking to apply this combination where signal integrity and model reliability matter equally.

EDUCATION

Istanbul Medipol University
Bachelor's Degree, Electrical and Electronics Engineering (3rd Year Student)

Expected Graduation: June 2028

PROFESSIONAL EXPERIENCE

Ziraat Teknoloji A.Ş.
System and Virtualization Engineering Intern (Corporate IT & Operations)

July 2026 – August 2026

- Monitored and managed enterprise-scale IT infrastructure, VMware vSphere and Workstation virtualization architectures, and HPE iLO for out-of-band server management.
- Developed and deployed system automation workflows on Red Hat Enterprise Linux servers, and analyzed planned and unplanned downtime through root cause analysis alongside experienced engineering teams.

Gülverdi Elektrik
Electrical Engineering Intern (Field & Maintenance Operations)

August 2025 – September 2025

- Assisted in the maintenance and troubleshooting of electrical panels and industrial equipment, gaining hands-on experience in field operations and safety protocols.
- Supported tracking of technical inventory and equipment condition, building a working understanding of physical production environments.

KEY ENGINEERING & AI PROJECTS

IQForge — SDR Signal Datasets for Machine Learning
Python, PyTorch, SigMF · github.com/emrefbulut/iqforge

2026

- Built an open-source tool converting SDR captures into PyTorch datasets, reading SigMF natively (cf32_le, ci16_le, ci8, memory-mapped for large files).
- Eliminated a train/test leakage failure mode: neighbouring windows share samples, so naive window-level splitting inflated accuracy by up to 13.6 points. Splitting at the recording level removes it. Added nuisance-variable balancing across SigMF metadata, and made the tool halt with a diagnostic error rather than silently degrade.

RoomGate-AI — Smart Access & Occupancy Monitor
Python, YOLO26, Edge Computing · github.com/emrefbulut/RoomGate-AI

May 2026 – June 2026

- Engineered an AI-driven access-control prototype bridging computer vision with hardware: an edge-optimized YOLO26 detection model driving a relay-based lock.
- Implemented spatial region-of-interest filtering so detections only trigger inside defined zones, cutting false actuations.

VoltPilot — EV Grid Readiness Simulation
TypeScript, Power Systems Modelling · github.com/emrefbulut/VoltPilot

April 2026 – June 2026

- Developed a data-driven simulation that analyzes a facility's electrical grid capacity before EV charging infrastructure investment, covering transformer monitoring, battery management, and telemetry validation.

LEADERSHIP & EXTRACURRICULAR

IEEE Communications Society (ComSoc) — Chairman
Istanbul Medipol Uni.

Coordinated multidisciplinary engineering teams, organized technical seminars, and built a culture of continuous learning across the student branch.

Sirada — Co-Founder
Dec 2025 – July 2026

Led end-to-end product development of a B2B smart queue and reservation platform, applying agile methods to iterate on features from performance data and user feedback.

SKILLS & COMPETENCIES

RF & Signal Processing: Software-Defined Radio (SDR), SigMF, IQ data handling, windowing and normalization, RF dataset construction, 5G application concepts

Machine Learning & Computer Vision: PyTorch, object detection (YOLO26, YOLOv8, MobileNetV3), model training and fine-tuning, dataset design, data labeling, train/test leakage prevention, edge deployment

Programming: Python, Java, C

Systems & Infrastructure: VMware vSphere/Workstation, Red Hat Enterprise Linux, HPE iLO, IT automation, root cause analysis, MQTT, REST APIs, SQLite

Soft Skills: Analytical problem-solving, initiative, collaborative teamwork, adaptability

Languages: Turkish (Native), English (Advanced)